Take-Home Assessment

Incoming Lead API + Lead Management Mini System

This assessment evaluates practical system design, API integration, database thinking, and implementation quality through a small real-world lead intake workflow.

Assessment Scenario

Your system should receive incoming lead data from an external source, store it in Supabase, and provide a simple internal interface for users to view and manage those leads.

Objective

- Design a small but practical workflow for receiving and processing incoming lead data.
- Implement a working API endpoint and persist relevant records in Supabase.
- Build a simple interface for lead viewing, status updates, assignment, and notes.
- Design a clear assignment flow that supports round-robin logic, agent online/offline availability, and agent notifications.
- Document your thinking clearly so the system can be reviewed and discussed.

Scope of Work

1. Incoming Lead API

Create an API endpoint to accept incoming lead data in JSON format.

Suggested endpoint	POST /api/leads/incoming

Example payload

Field	Value
leadId	"LD1001"
name	"John Tan"
phone	"0123456789"
email	"john@email.com"
source	"Facebook Ads"
project	"Residensi Mutiara"
budget	650000
message	"Interested in 3-bedroom unit"
createdAt	"2026-04-20T10:30:00Z"

- validate the incoming payload and return a clear API response

- authenticate or validate incoming requests using a simple token or similar mechanism
- detect and handle duplicate leads based on phone number and/or email
- store the incoming lead and log the raw payload or event for audit/reference
- handle invalid or incomplete data gracefully

2. Database

Use Supabase as the database. The structure may be designed based on your own assumptions, but it should be logical, practical, and clearly explained.

- lead information
- assigned agent information
- agent online/offline availability or presence status
- notes and lead status history
- lead assignment history and assignment sequence for audit or troubleshooting
- raw incoming payload logs or webhook event logs
- notification records sent to agents

3. Lead Management Interface

Provide a simple user interface for internal users. The UI does not need to be highly polished, but it should be functional and easy to understand.

- View a list of leads.
- View lead details.
- Update lead status.
- Assign a lead to an agent.
- Add notes to a lead.
- Show agent online/offline status in the UI.

4. Assignment and Notification Logic

Design and implement an agent assignment flow that reflects a simple real-world lead distribution scenario.
The logic should:

- follow round-robin order across eligible online agents
- skip offline agents automatically
- keep track of assignment order so the next lead goes to the next eligible agent

- include agents again when they come back online
- continue assignment using remaining online agents when one goes offline
- create a notification record when a lead is assigned to an agent

Example assignment behavior

If agents A, B, and C are online, the first three leads should be assigned to A, then B, then C. If agent D later comes online, the next lead should go to D. If B becomes offline, the next assignment cycle should continue using only A, C, and D.

5. Documentation

- Flowchart of the overall process.
- Database schema or ERD.
- API endpoint list.
- Setup instructions.
- Assumptions made and design decisions.
- Brief explanation of the round-robin assignment approach and presence-tracking logic.

Testing Method

A real external sender is not required for this assessment. You may test the incoming API locally using Postman, curl, Insomnia, or a similar tool. You may simulate agent online/offline status and assignment outcomes using seeded test data, simple admin actions, or a local script. Please include clear testing steps in the README.

```
Example curl command
curl -X POST http://localhost:3000/api/leads/incoming \
  -H "Content-Type: application/json" \
  -d '{ ...sample payload... }'
```

Deliverables

- Source code
- README
- Database schema
- Flowchart
- Short architecture or design explanation

- Brief test notes or sample steps showing how round-robin assignment and online/offline behaviour were verified

Bonus Considerations

If time permits, candidates may extend the solution with additional features or technical enhancements that they believe would add practical value to the system. These may include improvements related to data validation, security, retry handling, dashboards, search, real-time communication, agent presence tracking, notifications, workload visibility, audit logging, caching, queue-based processing, or assistant-style operational features. Candidates are not expected to implement all of these. We will value thoughtful prioritization, sound technical judgment, and a clear explanation of why particular enhancements were chosen.

Candidate Guidance

You are not expected to build a production-ready system. Focus on delivering a clean, working solution with sensible structure, clear documentation, and reasonable assumptions. A simple but clearly explained implementation is better than an overly large solution.

End of Assessment